

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	xxxxxx
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve students' learning experiences by integrating Artificial Intelligence and Natural Language Processing to identify emotions from text and generate personalized learning guidance. The system combines deep learning models (BERT and BiLSTM) with Google's Gemini API to detect emotions, provide empathetic responses, and visualize learning analytics through an interactive dashboard.

Project Overview	
Objective	The primary objective is to develop an AI-powered learning support system capable of detecting students' emotions from textual input and generating personalized learning recommendations. The platform aims to improve engagement, confidence, and academic performance through intelligent emotional analysis.
Problem Statement	
Description	Students often experience emotions such as confusion, frustration, boredom, or anxiety while learning. Existing educational platforms provide generic assistance but cannot recognize students' emotional states or personalize learning support accordingly.
Impact	Providing emotion-aware learning guidance can improve student engagement, increase learning efficiency, reduce frustration, and create a more personalized educational experience.

Proposed Solution	
Approach	Develop an intelligent emotion detection platform that analyzes student learning problems using deep learning models (BERT and BiLSTM), identifies primary and mixed emotions, and generates personalized learning guidance using Google's Gemini API.
Key Features	☐ AI-based emotion detection using BERT and BiLSTM models. ☐ Mixed-emotion detection with confidence score calculation. ☐ Personalized learning recommendations using Gemini API. ☐ Emotion history storage using SQLite and CSV logs. ☐ Interactive Streamlit dashboard with analytics and visualization. ☐ Real-time emotion prediction and guidance generation.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5/i7 Processor or equivalent
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB Free Disk Space (SSD Recommended)
Software		
Frameworks	Python frameworks	Streamlit
Libraries	Additional libraries	Python 3.12
Development Environment	IDE	PyTorch, Transformers, Pandas, NumPy, Scikit-learn, Plotly, Matplotlib
Data		
Data	Source, size, format	SQLite Database, CSV Files